

“Analyzing E-Learning Platform Purchases using MySQL”

Project Overview

This project analysed an online learning platform's purchase data using MySQL. The objective was to understand learner purchasing behaviour, identify popular course categories, sales trend, and generate actionable business insights using SQL queries, joins, subqueries, Common Table Expressions (CTEs), views, and aggregate functions.

Key Insights

- The **Programming** category generated the highest revenue due to its higher course prices and multiple purchases.
- **Python for Data Science** emerged as the most purchased course based on total quantity sold.
- A small number of learners contributed a larger share of the overall revenue, while most learners fell into the **Low Value** spending category.
- Learners purchasing courses from multiple categories showed greater engagement with the platform compared to those purchasing from only one category.
- The **category performance view** provided a consolidated summary of revenue, purchase count, and average revenue per purchase for each course category, making reporting more efficient.

Recommendations

- Focus marketing on **top-performing course categories**, especially Programming, to maximize revenue.
- Recommend related courses to learners who have already completed one course to encourage cross-category purchases.
- Improve data quality by reducing missing (NULL) values in important fields such as course name, category, and price.
- Introduce offers or discounts for low-performing categories to increase enrolments.
- Regularly monitor category performance using SQL views and dashboards to support business decision-making.

Conclusion

The SQL analysis demonstrated how relational database concepts, joins, aggregate functions, subqueries, CTEs, views, and NULL handling can be used to extract meaningful business insights from an online learning platform. The findings can help management optimize marketing strategies, improve course offerings, enhance learner engagement, and increase overall revenue.